

● HARD

● ALGEBRA

Algebra

10 questions · ~15 min

10

Q1

GRID-IN ALGEBRA

The graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. What is the x -coordinate of the x -intercept of the resulting graph?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

ALGEBRA

For the function f , if $f(3x) = x - 6$ for all values of x , what is the value of $f(6)$?

- A. -6
- B. -4
- C. 0
- D. 2

Q3

ALGEBRA

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 12 and the fourth odd integer is at most 26 less than the sum of the first and third odd integers. Which inequality represents this situation?

- A. $12x + 6 \leq x + x + 4 - 26$
- B. $12x + 6 \geq 26 - x + x + 4$
- C. $12x + 4 \leq x + x + 3 - 26$
- D. $12x + 4 \geq 26 - x + x + 3$

Q4

GRID-IN

ALGEBRA

What value of t is the solution to the equation $0.8t - 0.46 = 8t - 0.001 + 1.9$?

STUDENT-PRODUCED RESPONSE

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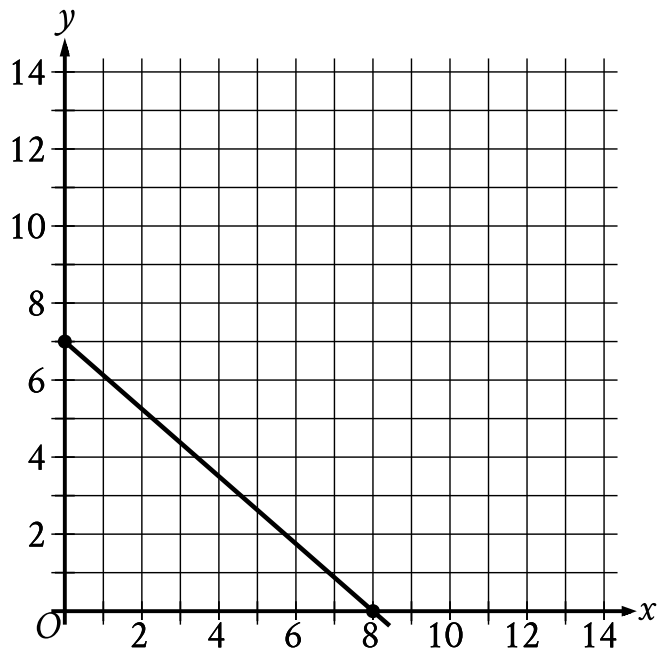
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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

One of the two equations in a linear system is $2x + 6y = 10$. The system has no solution. Which of the following could be the other equation in the system?

- A. $x + 3y = 5$
- B. $x + 3y = -20$
- C. $6x - 2y = 0$
- D. $6x + 2y = 10$



- The line slants gradually down from left to right.
- The line passes through the following points:
 - (0 comma 7)
 - (8 comma 0)

The point with coordinates $(d, 4)$ lies on the line shown. What is the value of d ?

- A. $\frac{7}{2}$
- B. $\frac{26}{7}$
- C. $\frac{24}{7}$
- D. $\frac{27}{8}$

x	-11	-10	-9	-8
f(x)	21	18	15	12

The table above shows some values of x and their corresponding values $f(x)$ for the linear function f . What is the x -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(-3,0)$
- B. $(-4,0)$
- C. $(-9,0)$
- D. $(-12,0)$

A small business owner budgets \$ 2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays \$ 4.90 per candle to purchase small candles and \$ 11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

ALGEBRA

$$5y = 10x + 11$$

$$-5y = 5x - 21$$

The solution to the given system of equations is x, y . What is the value of $30x$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A certain apprentice has enrolled in 85 hours of training courses. The equation $10x + 15y = 85$ represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.